Name: ______________________________ Date: _______

Yr 8 Mathematics (General & Ext)

Investigation, 2011

Topic – Algebra (In Class Validation)

___ / 20
=___ %

Weighting: *5% of the year*

Aim

Use algebra to investigate a formula.

The investigation

This is an investigation called 'Opposite Corners'. You will be investigating a feature of the 15 x 15 grid of numbers, using multiplication and subtraction. Follow the instructions and answer the questions in the spaces provided. You can draw squares and rectangles onto this grid for your investigation.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
211	212	213	214	215	216	217	218	219	220	221	222	223	224	225

1. [2 marks]

A 2x2 square is drawn on to the 15x15 Square.

There are two sets of opposite corners. In the square highlighted the diagonally opposite corners of the square are 1 and 17 (one set) and 2 and 16 (second set). Multiply the opposite corners and find the difference between the answers. i.e.

Do 1x17 and 2x16, then subtract the lowest answer from the highest. This is the difference. (If you use a calculator for the multiplication, have a go at the subtraction to get some marks!)

1x17 = 17

2x16 = 32

Difference 15

$$\begin{array}{r} 32 \\ -17 \\ \hline 15 \end{array}$$

2. [2 marks]

Now draw another 2x2 square onto the big grid and repeat the process: multiply the opposite corners and find the difference between the answers.

225 × 209 = 47025
210 × 224 = 47040
15

3. [1 mark]

What do you notice about your answers? Do you think this will always happen?

Yes. Answers will be the same for same size squares as it was on the 100 chart (10x10.) The first difference then for a 2x2 was 10. Here, the first difference for a 2x2 is 15. Because it is a 15 x 15 or 225 chart.

4. [2 marks]

Repeat the processes with a 3x3 square i.e. draw a 3x3 square onto the grid (as highlighted in the example below), multiply the opposite corners and find the difference between the answers. Show working out for two different squares.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
211	212	213	214	215	216	217	218	219	220	221	222	223	224	225

1st square

52 × 84 =
54 × 82 =

2nd square

5. [2 marks]

Continue to investigate the differences between the opposite corners of squares of sizes 4x4 (Make sure you show working out for two different squares)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
211	212	213	214	215	216	217	218	219	220	221	222	223	224	225

1st square

1 x 49 = 49
4 x 46 = 184
135

2nd square

177 x 225 = 39825
180 x 222 = 39960
135

6. [2 marks]

Continue to investigate the differences between the opposite corners of squares of sizes 5x5 (show working out for two different squares)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
211	212	213	214	215	216	217	218	219	220	221	222	223	224	225

1st square

61 x 5 = 305
1 x 65 = 65
240

2nd square

11 x 75 = 825
71 x 15 = 1065
240

7. [3 marks]
Complete the following table with your results.

Square size (n)	Difference (D)
2 x 2	15
3 x 3	60 ✓
4 x 4	135
5 x 5	240 ✓
N x N	$15(n-1)^2$ ✓

8. [2 marks]
Do you notice anything about your results? Are they related to a special set of numbers? Are they all multiples of the same number?

they are all multiples of 15 (5 and 3 also.)
15 as the HCF.
✓

9. [2 marks]

(any other legitimate answer). ✓

As you can see, there is a column in the table for you to enter the formula for the difference, D, for a general square of side length N. Write a formula below to find D given N.

✓ $D = 15(n-1)^2$ ✓

10. [2 marks]
Use your formula to predict the difference in the square size 8 x 8. Show working out.

$D = 15(8-1)^2$ ✓

$= 15(7)^2$

$= 15 \times 49$

$= 735$ ✓